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(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Prof. Riddhi Doshi
2)Arpita Waghmare
3)Sampada Bandre
4)Lavanya Tiwade

(57) Abstract :
ABSTRACT FORENSIC FACIAL IMAGE GENERATION AND CRIMINAL IDENTIFICATION SYSTEM USING LATENT DIFFUSION MODELS The present invention discloses a forensic facial image generation and criminal identification system (100) using latent diffusion models. The system comprises a text processing module (110) employing a CLIP text encoder (114) for converting natural language witness descriptions into embedding tensors, an image generation module (120) utilizing a latent diffusion model with a U-Net denoiser (124) operating in a compressed latent space for computationally efficient image synthesis, a dual-mode prompt engineering module (130) for simultaneously generating photorealistic mugshots and forensic pencil sketches with structured negative prompts, a face matching module (140) employing an ensemble of deep learning face recognition models (142, 144) with cosine similarity scoring for biometric comparison against a criminal database, and a database management module (150) with session persistence. The system achieves a 48-fold computational reduction over pixel-space diffusion, produces dual forensic outputs from a single text input, and enables automated suspect identification through calibrated similarity thresholds. [to be published with fig.1]

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